

Chapter 6a:

Application of Newton's Laws (1)

Learning Objectives

6.1 Solving Problems using Newton's Laws

6.2 Force of friction

Drawing a Free Body Diagram

1. **Identify all forces acting on the object.**
2. **Draw a coordinate system.** Use the axes defined in your pictorial representation. If the axes are tilted for motion along an incline, then the axes of the free-body diagram should be similarly tilted.
3. **Represent the object as a dot at the origin of the coordinate axes.** This is the particle model.
4. **Draw vectors representing each of the identified forces.** Be sure to label each force vector.
5. **Draw and label the *net force* vector,** which points in the same direction as the *acceleration* vector on your motion diagram.

Things to remember...

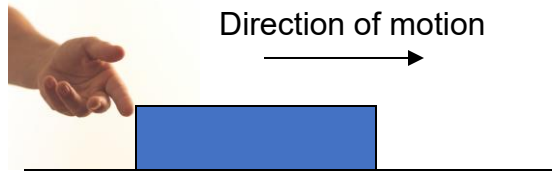
- **Free body diagrams**
- **Newton's 2nd Law: $F_{\text{net}} = \sum F = ma$**
- **Mass $\neq$ Weight**

This week is all about learning how to find specific forces and then adding them together to equal “ma”

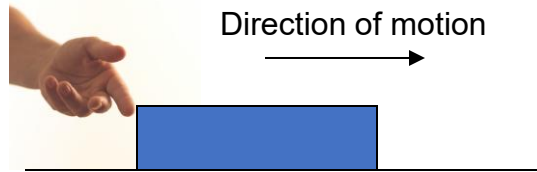
Common Forces and Notations

Force	Notation
General force	$\vec{\mathbf{F}}$
Gravitational force	$\vec{\mathbf{F}}_{\mathbf{g}}$ or $\vec{\mathbf{W}}$
Spring force	$\vec{\mathbf{F}}_{\mathbf{s}}$
Tension	$\vec{\mathbf{T}}$
Normal force	$\vec{\mathbf{N}}$
Static friction	$\vec{\mathbf{f}}_{\mathbf{s}}$
Kinetic friction	$\vec{\mathbf{f}}_{\mathbf{k}}$
Drag	$\vec{\mathbf{D}}$
Thrust	$\vec{\mathbf{F}}_{\text{thrust}}$

Drawing a Free Body Diagram



no friction



with friction

$\vec{N}$

$\vec{F}_g$

$\vec{N}$

$\vec{F}_g$

$\vec{F}$

$\vec{F}_g$

$\vec{f}$

$\vec{F}_g$

$$\sum F_y = N - F_g = 0$$

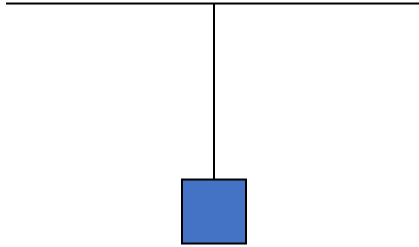
$$\sum F_y = N - F_g = 0$$

$$\sum F_x = F = m \cdot a_x$$

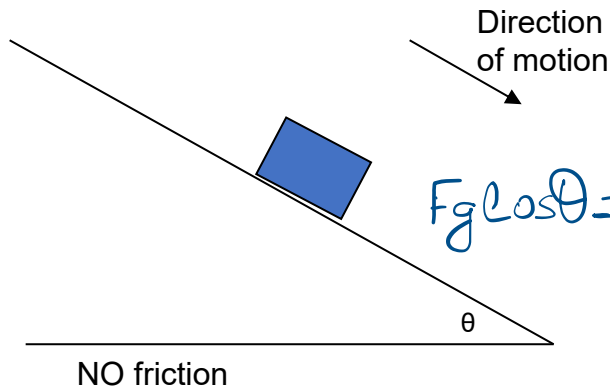
$$\sum F_y = N - F_g = 0$$

$$\sum F_x = F - f = m \cdot a_x$$

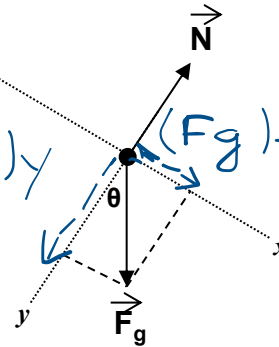
Drawing a Free Body Diagram



$$\sum F_y = T - F_g = 0$$

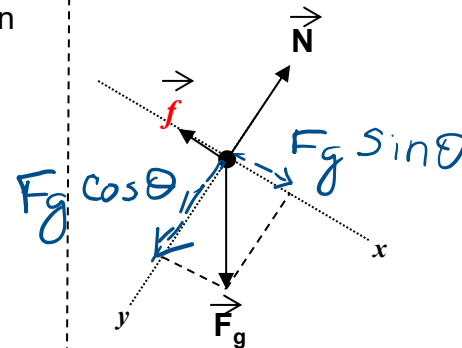
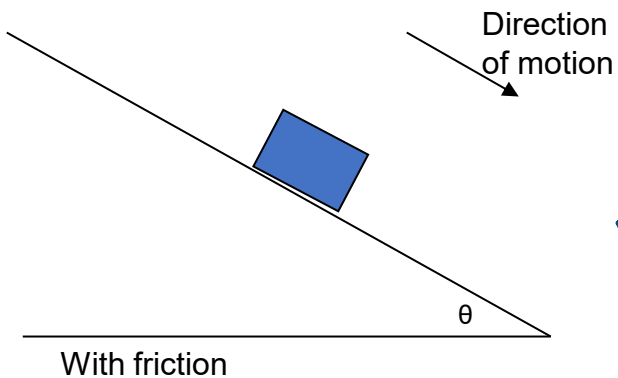


$$F_g \cos \theta = (F_g)_y$$



$$\sum F_y = N - F_g \cos \theta = 0$$

$$\sum F_x = F_g \sin \theta = m \cdot a_x$$



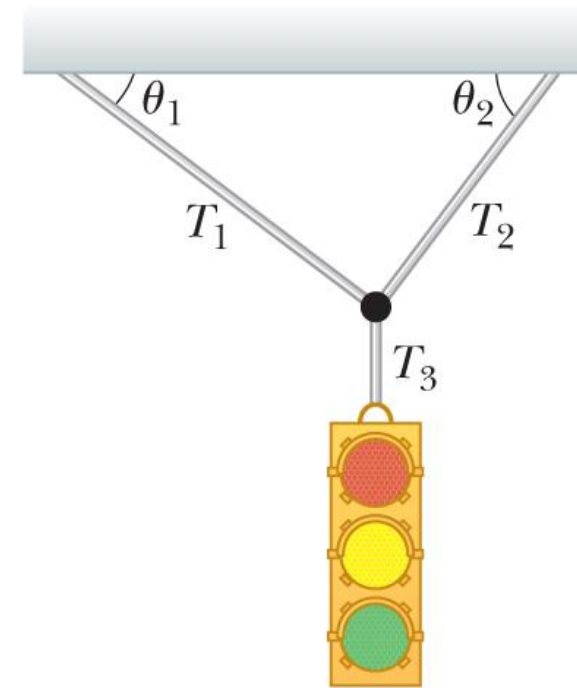
$$\sum F_y = N - F_g \cos \theta = 0$$

$$\sum F_x = F_g \sin \theta - f = m \cdot a_x$$

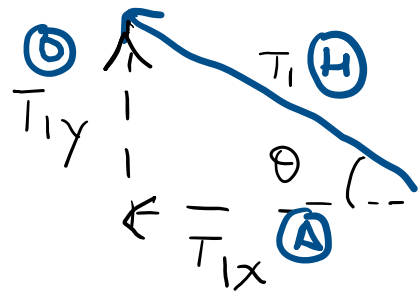
Example: Traffic Light

A traffic light weighing 122 N hangs from a cable tied to two other cables fastened to a support as the figure. The upper cables make angles of $\theta_1 = 37.0^\circ$ and $\theta_2 = 53.0^\circ$ with the horizontal.

These upper cables are not as strong as the vertical cable and will break if the tension in them **exceeds 100 N**. Does the traffic light remain hanging in this situation, or will one of the cables break?



Example: Traffic Light



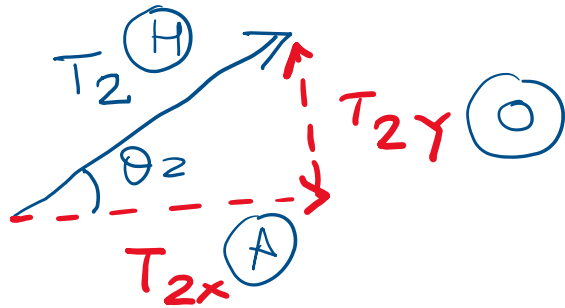
$$\sin \theta_1 = \frac{T_{1y}}{T_1}$$

$$T_{1y} = T_1 \sin \theta_1$$

$$\cos \theta = \frac{T_{1x}}{T_1}$$

$$T_{1x} = T_1 \cos \theta_1$$

Note: T_{1x} is in negative x-direction!

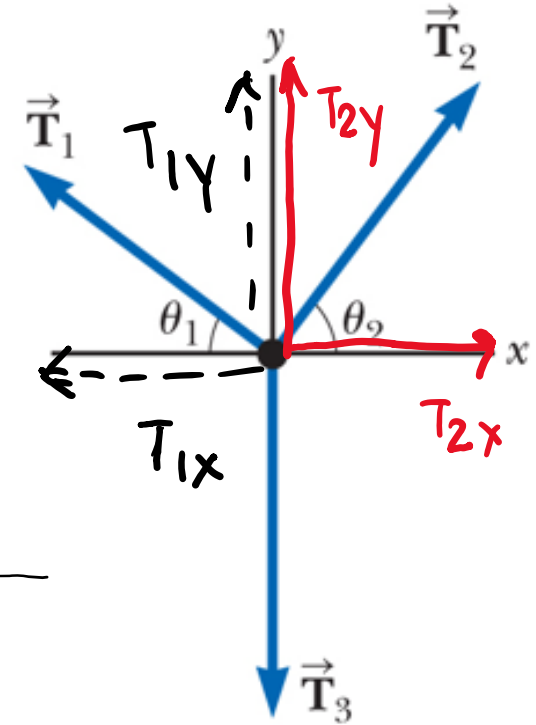


$$\sin \theta_2 = \frac{T_{2y}}{T_2}$$

$$T_{2y} = T_2 \sin \theta_2$$

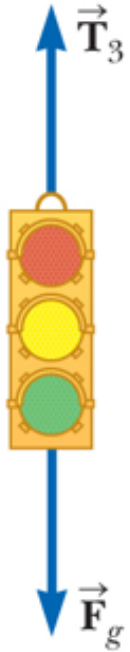
$$\cos \theta_2 = \frac{T_{2x}}{T_2}$$

$$T_{2x} = T_2 \cos \theta_2$$



Force	x Component	y Component
$\vec{T}_1$	$-T_1 \cos \theta_1$	$T_1 \sin \theta_1$
$\vec{T}_2$	$T_2 \cos \theta_2$	$T_2 \sin \theta_2$
$\vec{T}_3$	0	$-F_g$

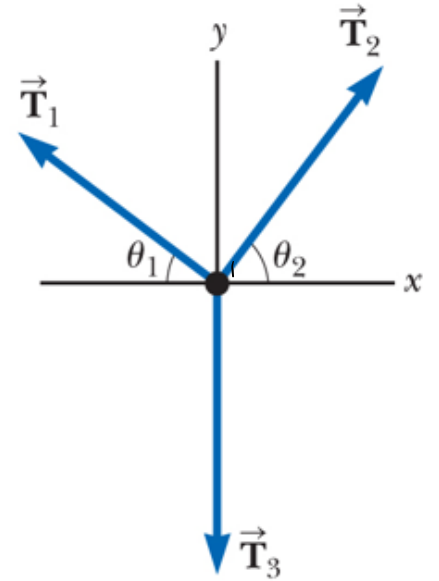
Example: Traffic Light



$$\sum F_y = 0$$

$$T_3 - F_g = 0$$

$$\Rightarrow T_3 = F_g = 122N$$



Force	x Component	y Component
$\vec{T}_1$	$-T_1 \cos \theta_1$	$T_1 \sin \theta_1$
$\vec{T}_2$	$T_2 \cos \theta_2$	$T_2 \sin \theta_2$
$\vec{T}_3$	0	$-F_g$

$$\sum F_x = -T_1 \cos \theta_1 + T_2 \cos \theta_2 = 0 \dots (1)$$

$$\sum F_y = T_1 \sin \theta_1 + T_2 \sin \theta_2 + (-F_g) = 0 \dots (2)$$

Example: Traffic Light

From equation (1) $-T_1 \cos \theta_1 + T_2 \cos \theta_2 = 0$

$$\Rightarrow T_2 = T_1 \left(\frac{\cos \theta_1}{\cos \theta_2} \right) \dots (3)$$

Substituting the value of T_2 from (3) into equation (2)

$$T_1 \sin \theta_1 + T_2 \sin \theta_2 + (-F_g) = 0$$

$$\Rightarrow T_1 \sin \theta_1 + T_1 \left(\frac{\cos \theta_1}{\cos \theta_2} \right) (\sin \theta_2) - F_g = 0$$

$$T_1 = \frac{F_g}{\sin \theta_1 + \cos \theta_1 \tan \theta_2}$$

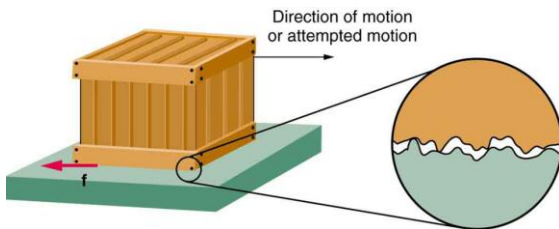
$$T_1 = \frac{122 \text{ N}}{\sin 37.0^\circ + \cos 37.0^\circ \tan 53.0^\circ} = \boxed{73.4 \text{ N}}$$

Substituting the value of T_1 into equation (3)

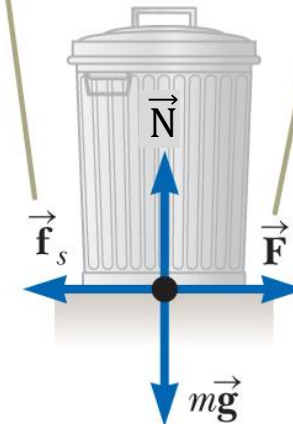
$$T_2 = (73.4 \text{ N}) \left(\frac{\cos 37.0^\circ}{\cos 53.0^\circ} \right) = \boxed{97.4 \text{ N}}$$

Force of Static Friction

- **Force of static friction f_s**
- As long as trash can not move $f_s = F$
 - If F increases $\rightarrow f_s$ also increases
 - If F decreases $\rightarrow f_s$ also decreases
- Friction force arises from nature of the two surfaces and is always parallel to the surface.
- Because of their roughness, contact made only at a few locations where peaks of material touch.

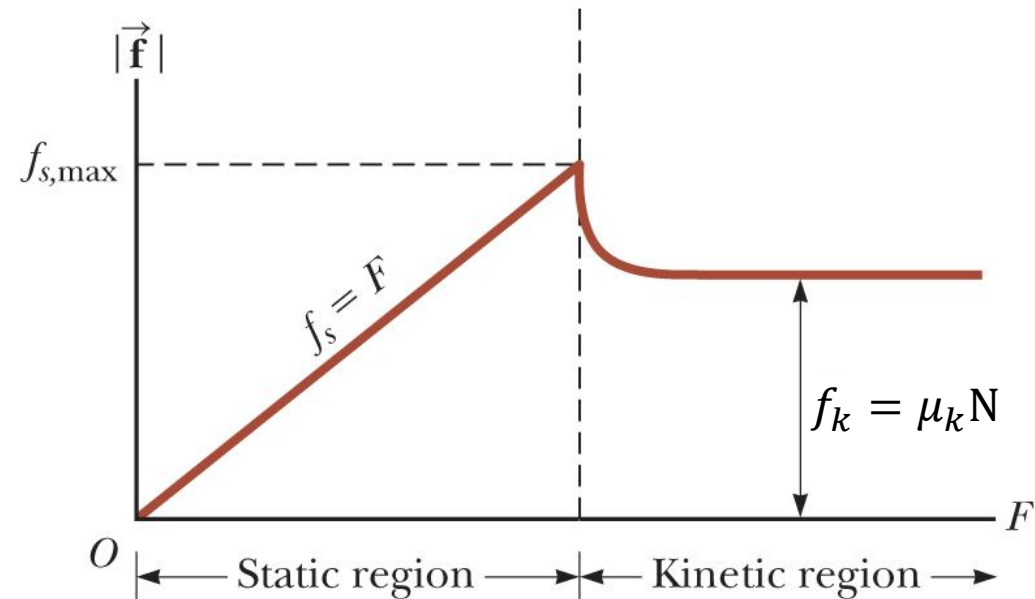
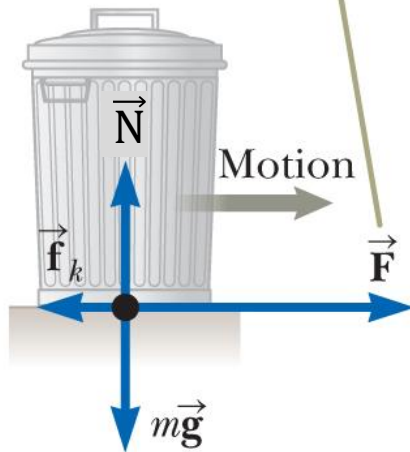


For small applied forces, the magnitude of the force of static friction equals the magnitude of the applied force.



Force of Kinetic Friction

When the magnitude of the applied force exceeds the magnitude of the maximum force of static friction, the trash can breaks free and accelerates to the right.



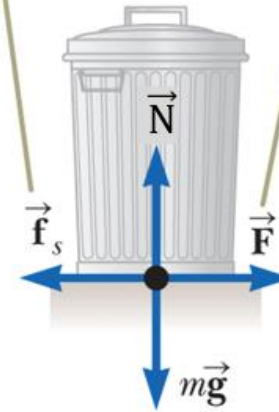
Two types of Frictional Forces

Force of static Friction, also called starting friction, is when the object is about to START moving.

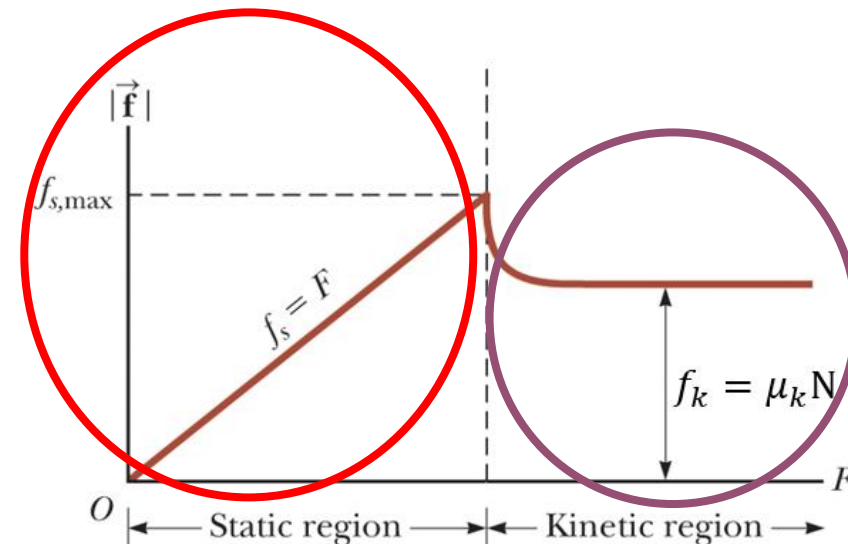
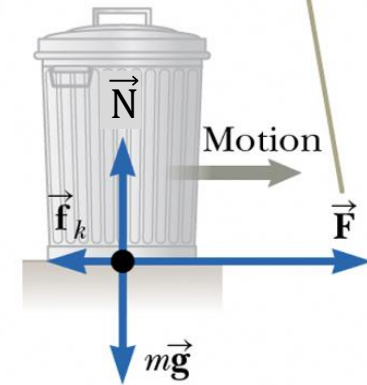
Force of kinetic friction, also called sliding friction, is when the object is MOVING.

$$f_k < f_s$$

For small applied forces, the magnitude of the force of static friction equals the magnitude of the applied force.



When the magnitude of the applied force exceeds the magnitude of the maximum force of static friction, the trash breaks free and accelerates to the right.



Coefficient of Friction

Static friction: $\vec{f}_s \leq \mu_s \vec{N}$ $\vec{f}_s = \vec{f}_{s,max} = \mu_s \vec{N}$

Kinetic friction: $\vec{f}_k = \mu_k \vec{N}$ $\vec{f} = \mu \vec{N}$

- $\mu_s =$ **coefficient** of static friction
- $\mu_k =$ **coefficient** of kinetic friction
- Values of μ_k and μ_s depend on nature of the surfaces, typical values: 0.03 to 1.0 $\rightarrow$ generally $\mu_k < \mu_s$
- **Direction** of friction force parallel to surface and opposite to the actual motion (kinetic friction) or the impending motion (static friction) of the object relative to the surface.

Coefficient of Friction

Coefficients of Friction

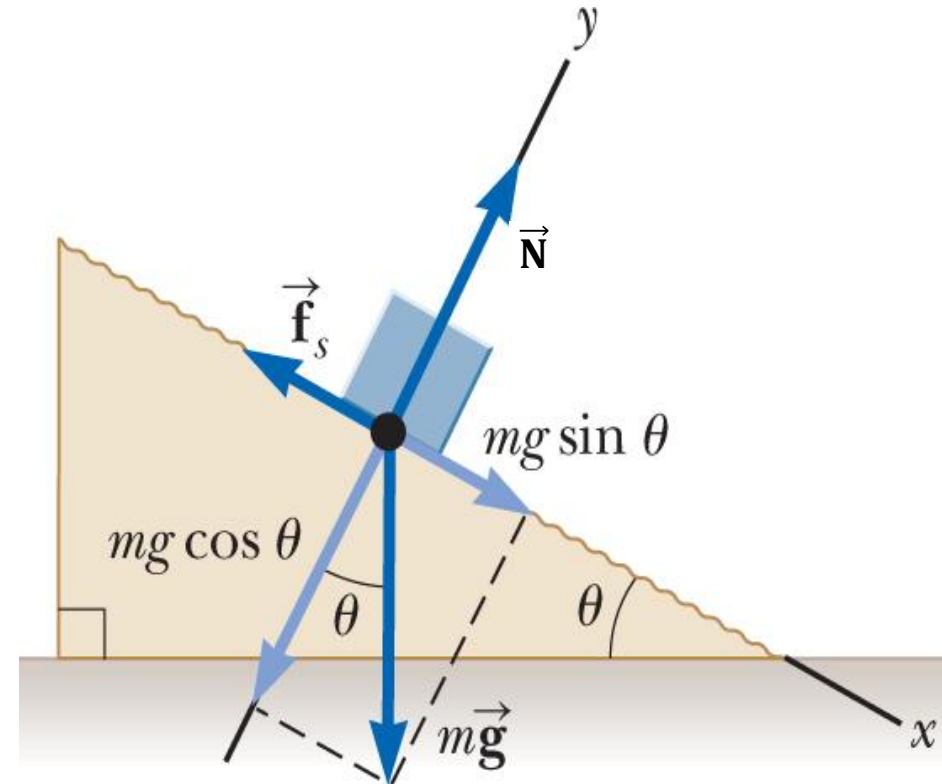
	μ_s	μ_k
Rubber on concrete	1.0	0.8
Steel on steel	0.74	0.57
Aluminum on steel	0.61	0.47
Glass on glass	0.94	0.4
Copper on steel	0.53	0.36
Wood on wood	0.25–0.5	0.2
Waxed wood on wet snow	0.14	0.1
Waxed wood on dry snow	—	0.04
Metal on metal (lubricated)	0.15	0.06
Teflon on Teflon	0.04	0.04
Ice on ice	0.1	0.03
Synovial joints in humans	0.01	0.003

Note: All values are approximate. In some cases, the coefficient of friction can exceed 1.0.

Experimental Determination of μ_s and μ_k

The following is a simple method of measuring coefficients of friction. Suppose a block is placed on a rough surface inclined relative to the horizontal as shown in the figure.

The incline angle is increased until the block starts to move. Show that you can obtain μ_s by measuring the critical angle θ_c at which this slipping just occurs.



Experimental Determination of μ_s and μ_k

$$\begin{aligned}\sum F_x &= mg \sin \theta - f_s = 0 \\ \sum F_y &= N - mg \cos \theta = 0 \\ f_s &= mg \sin \theta = \left(\frac{N}{\cos \theta} \right) \sin \theta = N \tan \theta\end{aligned}$$

$$\mu_s N = N \tan \theta_c$$

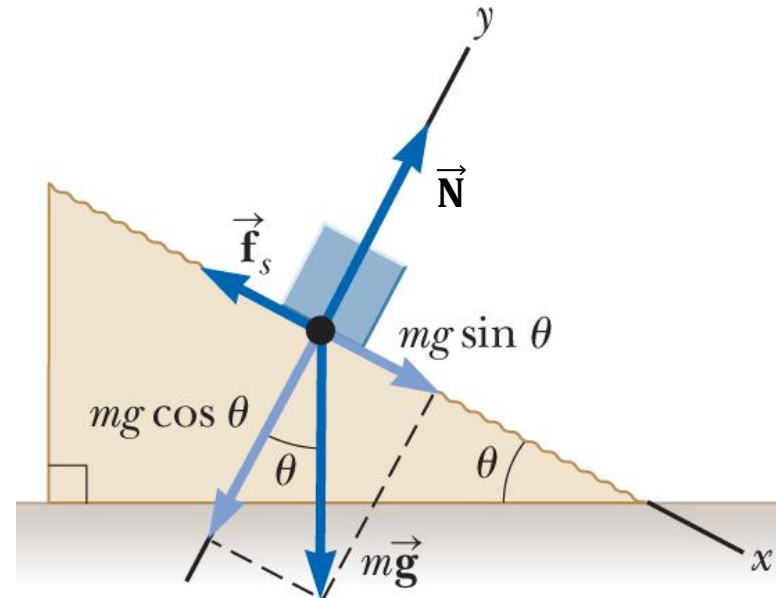
$$\mu_s = \tan \theta_c$$

$$N - mg \cos \theta = 0$$

$$N = mg \cos \theta$$

$$\frac{N}{\cos \theta} = mg$$

$$\vec{f}_s = \mu_s \vec{N}$$



ABCD card

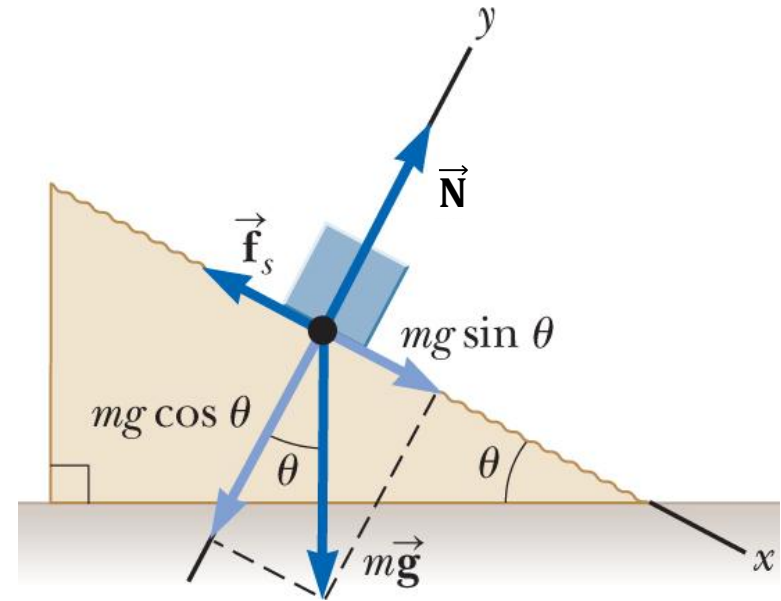
If the box just starts to slip at $\theta_c = 20.0^\circ$, what is the value of coefficient of **static friction** (μ_s)?

a. 0.364 N

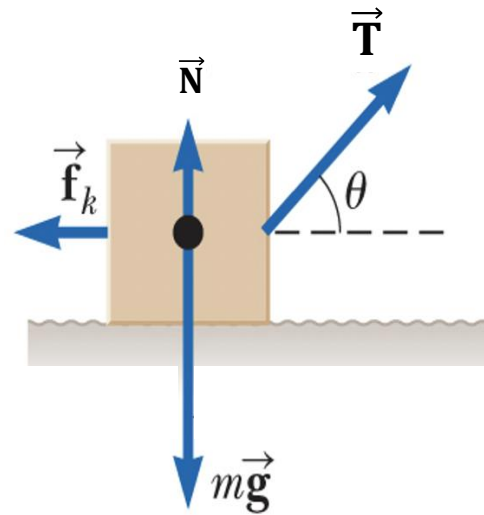
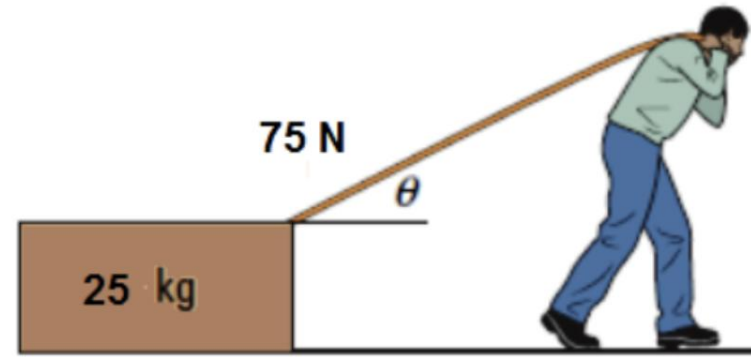
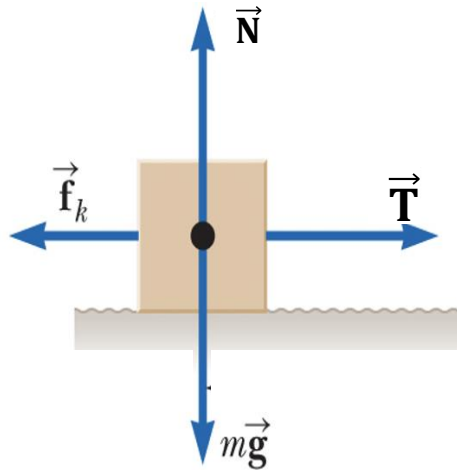
b. 0.364

c. 0.364°

d. 0 N



Example: Find the **acceleration**?



Example: Find the **acceleration**?

No **acceleration** along y direction,

$$\sum F_y = 0$$

$$N - F_g = 0$$

$$N = F_g = mg$$

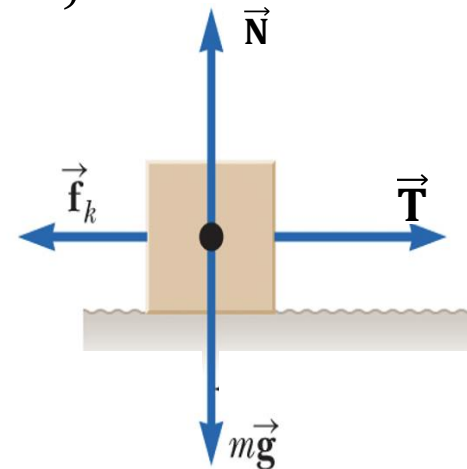


The object experiences **acceleration** along x direction,

$$\sum F_x = ma_x$$

$$T - f_k = ma_x \quad \vec{f}_k = \mu_k \vec{N}$$

$a_x = ??$ if μ_k and $\vec{T}$ is given



Example: Find the **acceleration**?

No **acceleration** along y direction,

$$\sum F_y = 0$$

$$N + T \sin \theta - F_g = 0$$

$$N = F_g - T \sin \theta$$

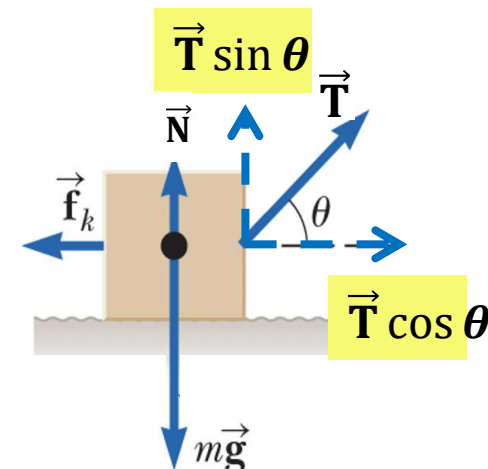
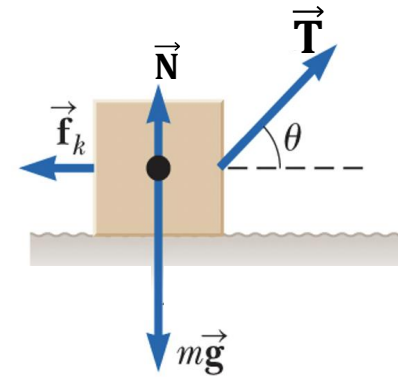
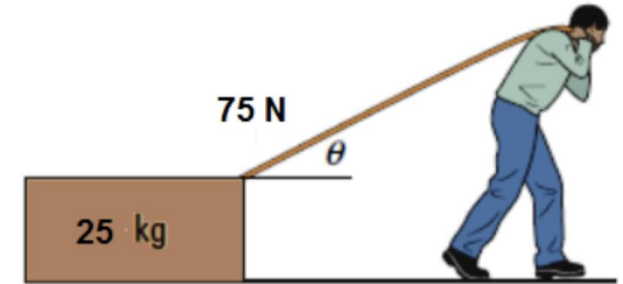
The object experiences **acceleration** along x direction,

$$\sum F_x = ma_x$$

$$T \cos \theta - f_k = ma_x$$

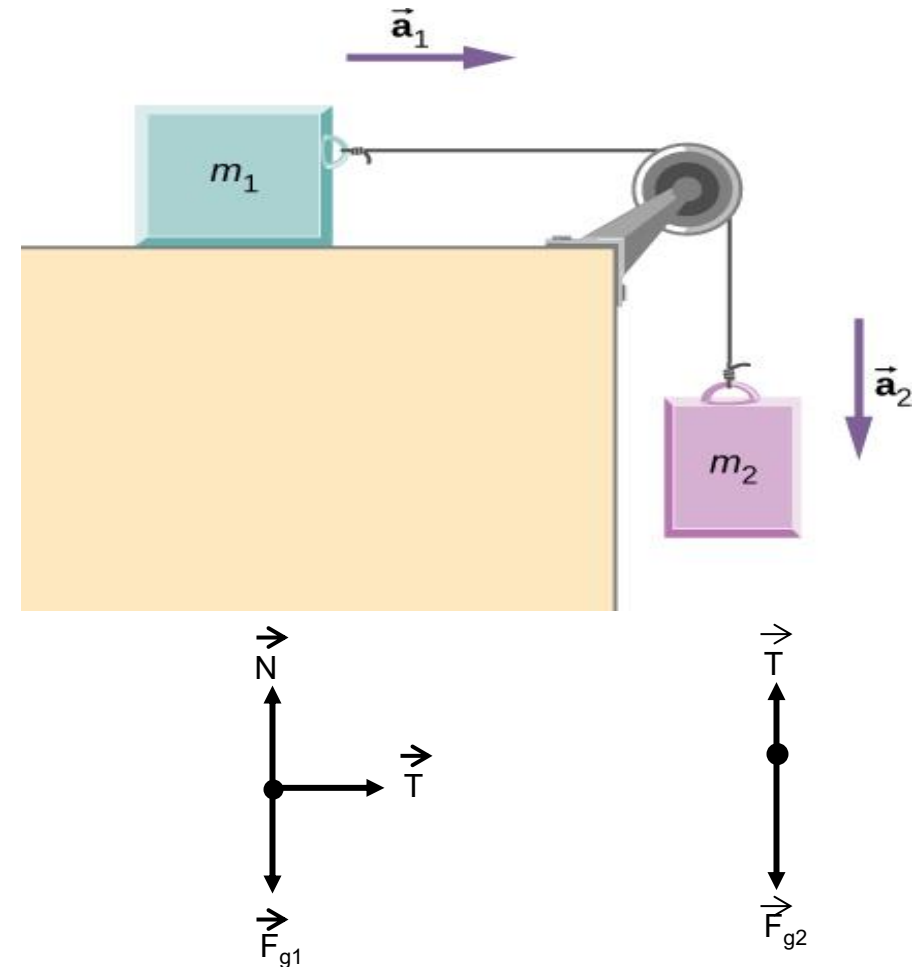
$$f_k = \mu_k \vec{N}$$

$$a_x = ?? \text{ if } \mu_k = 0.20$$

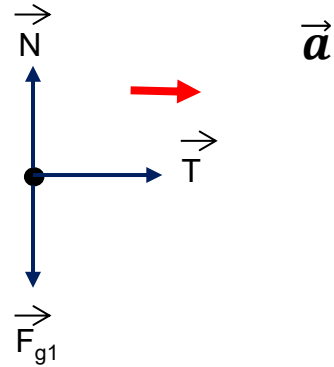


Problem Solving (Application of Newton's Law)

- i. Two blocks are connected by a massless rope as shown below. The mass of the block on the table is 4.00 kg and the hanging mass is 1.00 kg. The table and the pulley are **frictionless**.
- (a) Find the acceleration of the system.
- (b) Find the tension in the rope.
- (c) Find the speed with which the hanging mass hits the floor if it starts from rest and is initially located 1.00 m from the floor.



Problem Solving (Application of Newton's Law)



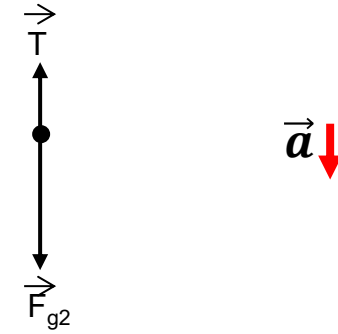
$$\sum F_y = 0 \text{ (no motion along } y - \text{axis)}$$

$$\vec{N} - \vec{F}_{g1} = 0$$

$$N = m_1 g \dots \dots (1)$$

$$\sum F_x = m_1 a_x$$

$$T = m_1 a \dots \dots (2)$$



$$\sum F_y = m_2 a_y$$

$$\vec{T} - \vec{F}_{g2} = m_2 (-a)$$

$$T - m_2 g = -m_2 a$$

$$T = m_2 g - m_2 a \dots \dots (3)$$

Problem Solving (Application of Newton's Law)

$$T = m_1 a \dots\dots(2)$$

$$T = m_2 g - m_2 a \dots\dots(3)$$

From equation (2) and (3)

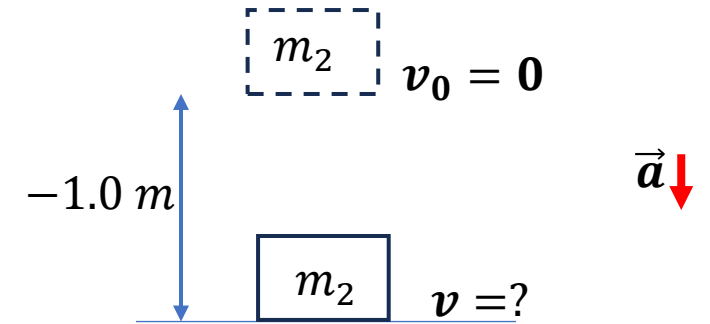
$$m_1 a = m_2 g - m_2 a$$

$$m_1 a + m_2 a = m_2 g$$

$$a(m_1 + m_2) = m_2 g$$

$$a = \frac{m_2 g}{(m_1 + m_2)} = 1.96 \text{ m/s}^2$$

From equation (2) $T = m_1 a = 7.84 \text{ N}$



Use kinematic equation to find speed

$$v^2 = v_0^2 + 2a(y - y_0)$$

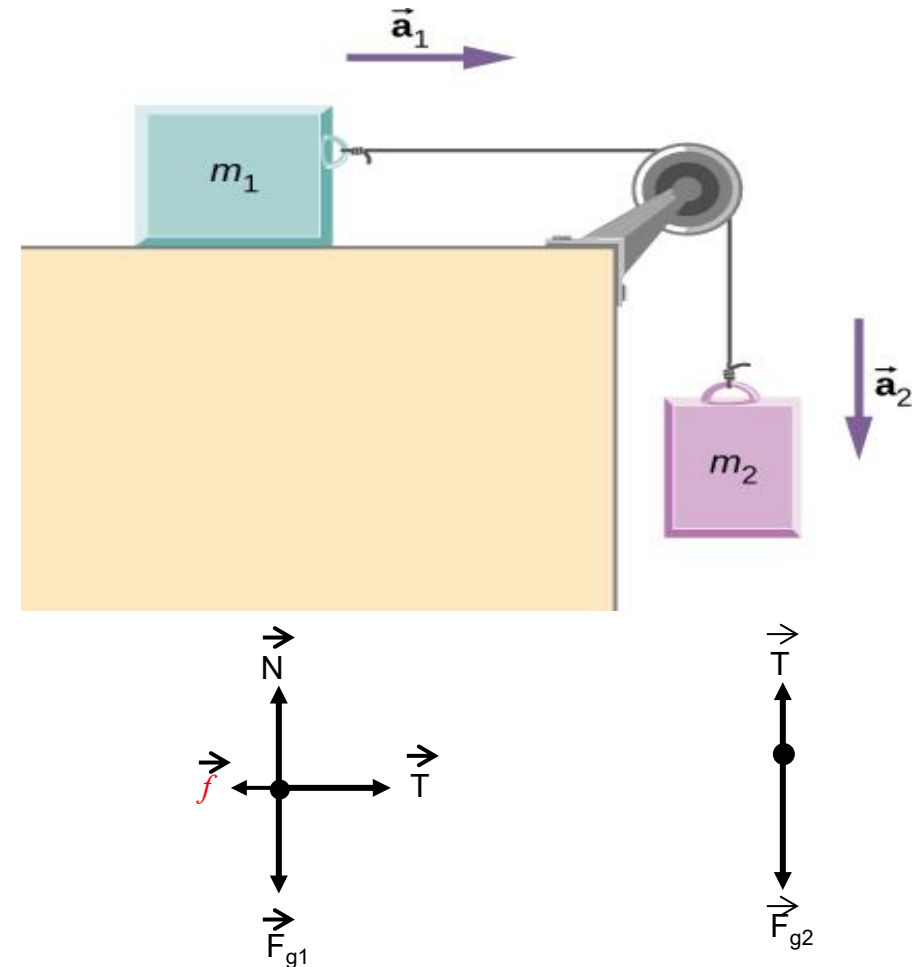
$$v^2 = 0 + 2(-1.96)(-1.00)$$

$$|v| = ?$$

Problem Solving (Application of Newton's Law)

ii. Two blocks are connected by a massless rope as shown below. The mass of the block on the table is 4.00 kg and the hanging mass is 1.00 kg. If there is a **coefficient of kinetic friction of 0.2** acting between the table and the mass m_1

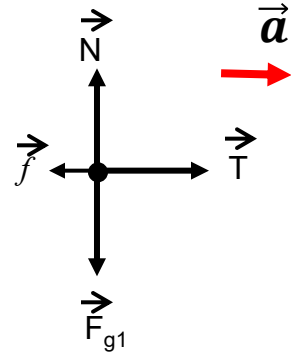
(a) Find the acceleration of the system.



$$\sum F_y = 0 \text{ (no motion along } y - \text{axis)}$$

$$\vec{N} - \vec{F}_{g1} = 0$$

$$N = m_1 g \dots \dots (1)$$



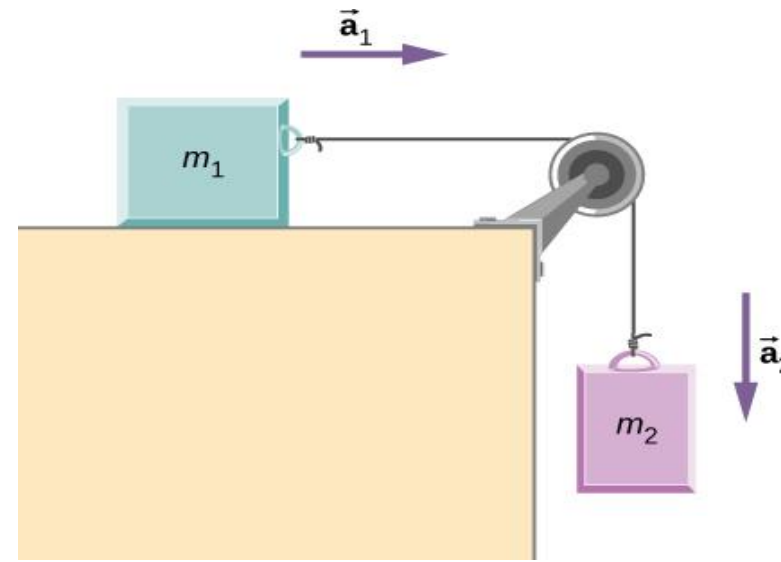
$$\sum F_x = m_1 a_x$$

$$T - f = m_1 a$$

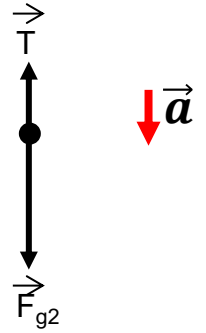
$$T - \mu_k N = m_1 a$$

$$T - \mu_k (m_1 g) = m_1 a \text{ (from equation 1)}$$

$$T = m_1 a + \mu_k (m_1 g) \dots \dots (2)$$



$$\vec{f}_k = \mu_k \vec{N}$$



$$\sum F_y = m_2 a_y$$

$$\vec{T} - \vec{F}_{g2} = m_2 (-a)$$

$$T - m_2 g = -m_2 a$$

$$T = m_2 g - m_2 a \dots \dots (3)$$

Problem Solving (Application of Newton's Law)

$$T = m_1 a + \mu_k(m_1 g) \dots \dots (2)$$

$$T = m_2 g - m_2 a \dots \dots (3)$$

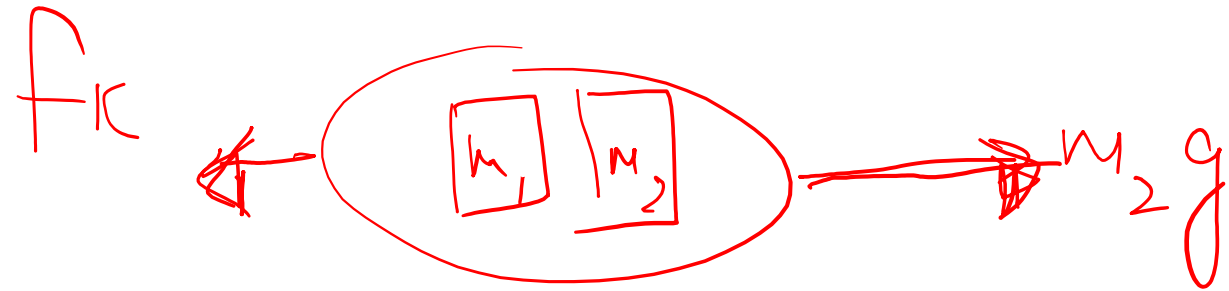
From equation (2) and (3)

$$m_1 a + \mu_k(m_1 g) = m_2 g - m_2 a$$

$$m_1 a + m_2 a = m_2 g - \mu_k(m_1 g)$$

$$a(m_1 + m_2) = m_2 g - \mu_k(m_1 g)$$

$$a = \frac{m_2 g - \mu_k(m_1 g)}{(m_1 + m_2)} = \dots \text{ m/s}^2$$



$$\Sigma F = m_T a$$

$$m_2 g - f_k = m_T a$$

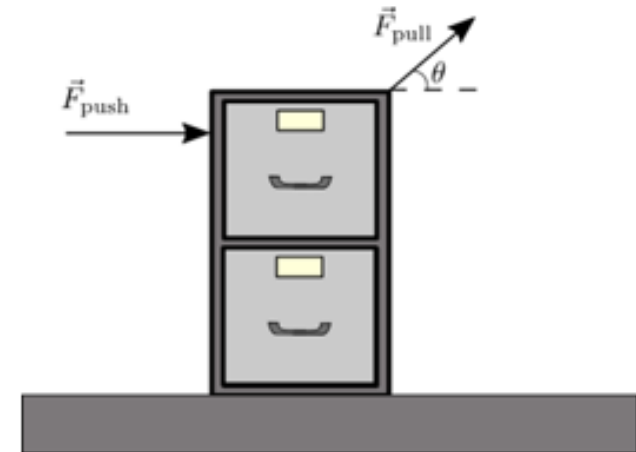
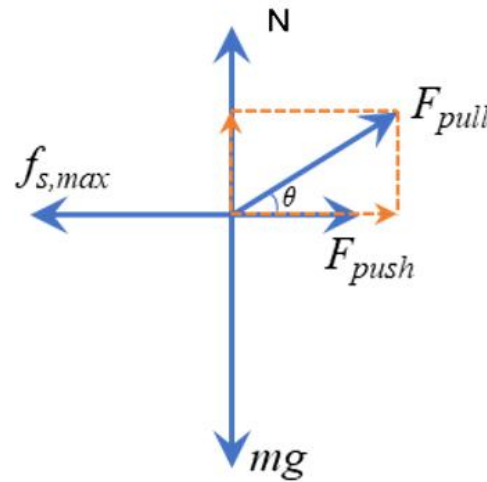
$$m_2 g - \mu_k N = (m_1 + m_2) a$$

$$\frac{m_2 g - \mu_k m_1 g}{(m_1 + m_2)} = a$$

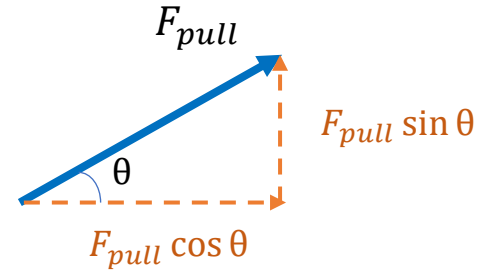
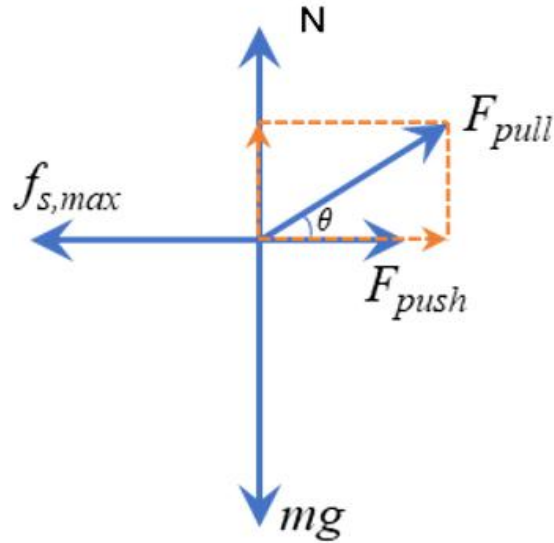
Problem Solving (Application of Newton's Law)

iii. A friend of yours is trying to pull a cabinet into her office. The cabinet has a mass of 75.0 kg, and she can't get it to budge on her own. You try to help and begin pushing horizontally in the same direction. Your friend is pulling with a force of 200 N at an angle $\theta = 25.0^\circ$ above the horizontal. If the coefficient of static friction between the cabinet and the floor is $\mu_s = 0.40$, **what force must you exert** ($\vec{F}_{push}$) to just get it moving?

Answer: 79.0 N



Problem Solving (Application of Newton's Law)



$$\vec{f}_s \leq \mu_s \vec{N}$$

$$\sum F_x = F_{push} + F_{pull} \cos \theta - f_{s,max} = 0$$

$$F_{push} = f_{s,max} - F_{pull} \cos \theta \dots (1)$$

From (1) and (2)

$$F_{push} = \mu_s (mg - F_{pull} \sin \theta) - F_{pull} \cos \theta$$

$$\sum F_y = N + F_{pull} \sin \theta - mg = 0$$

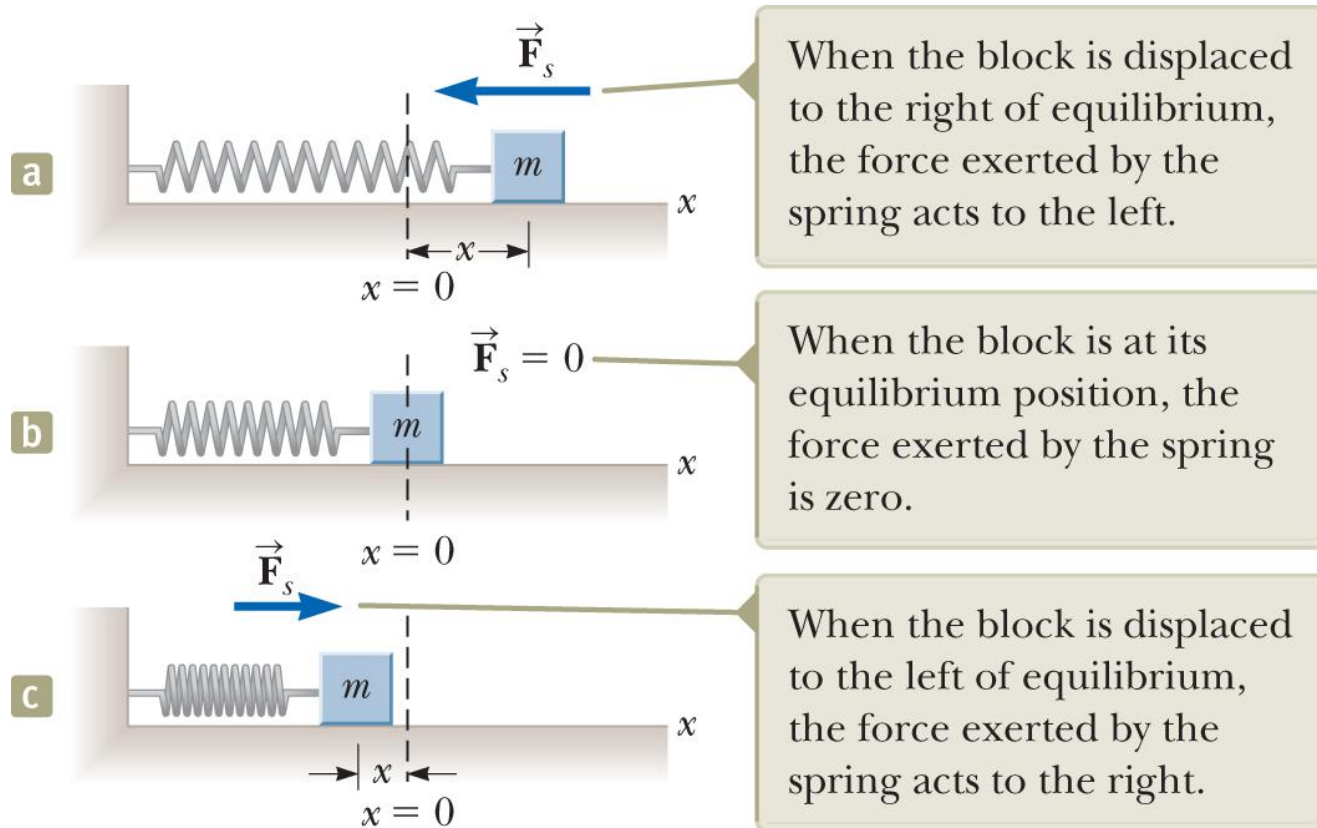
$$N = mg - F_{pull} \sin \theta$$

$$f_{s,max} = \mu_s N = \mu_s (mg - F_{pull} \sin \theta) \dots (2)$$

$F_{push} = \text{---} N$

Spring Force (recall...)

- Springs come in many forms. When deflected, they push or pull with a spring force.



Hooke's law:

$$F_s = -kx$$

F_s = spring force
 k = spring constant
 x = spring stretch or compression

$$F = -k(x - x_0)$$

$x_0 = 0$

Restoring force → always directed toward equilibrium position.

Problem Solving (Textbook)

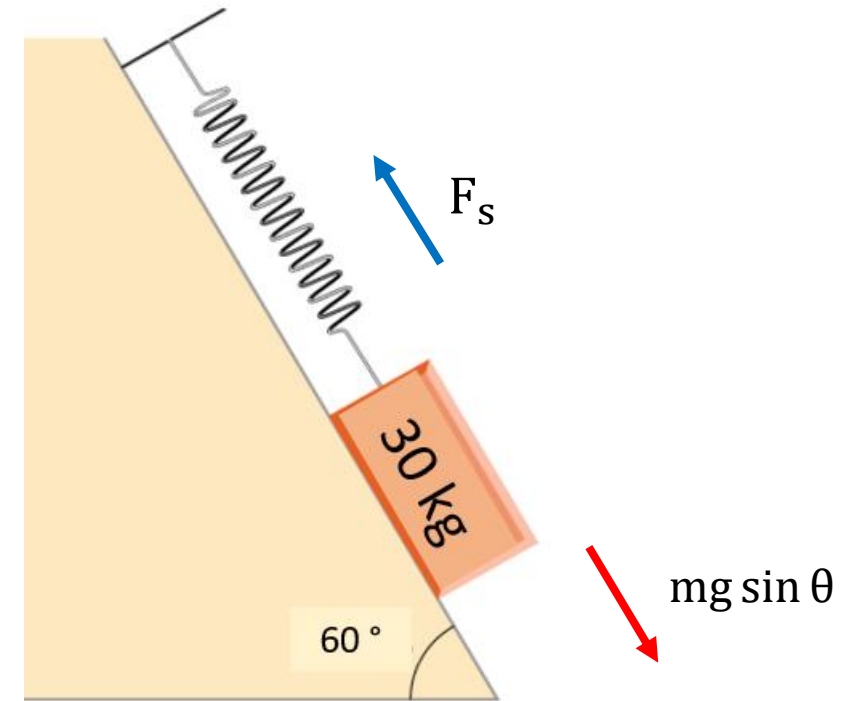
As shown in the figure a 30.0 kg block is resting on a frictionless ramp inclined at 60° to the horizontal. The block is held by a spring that is stretched 5.0 cm. What is the spring constant of the spring?

Hooke's law:

$$F_s = -kx$$

$$mg \sin \theta = kx$$

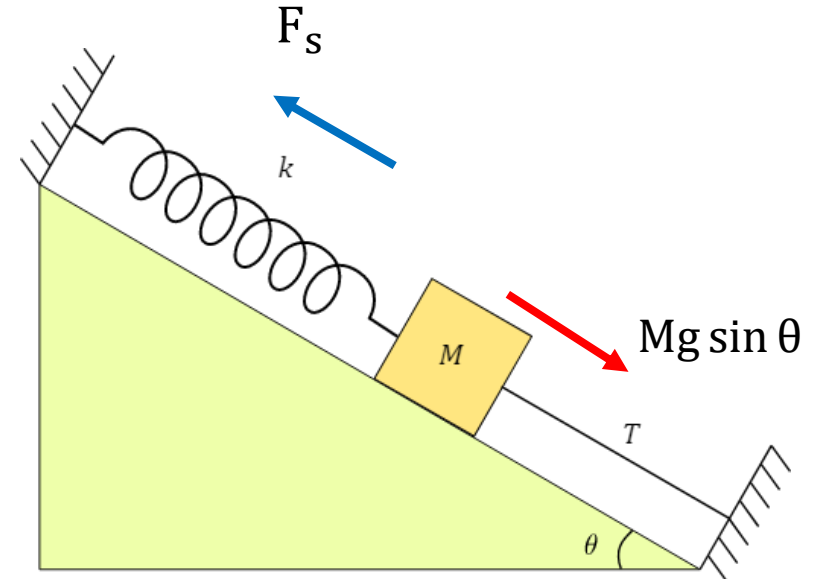
$$k \approx 5092 \text{ N/m}$$



Problem Solving #6 (Spring Force)

A block of mass $m = 10.0$ kg is at rest on a frictionless inclined plane with angle $\theta = 30^\circ$. It is attached to an extended spring with spring constant $k = 330$ N/m on the higher end of the inclined plane and with a taut rope with tension $T = 50.0$ N at the other end, as shown in the figure. How **long is the spring extended** long is the spring extended relative to its relaxed length?

Answer: 0.30 m



Problem Solving #6 (Spring Force)

$$\sum F_x = T + F_g \sin \theta - F_{sp} = 0$$

Spring force, $F_{sp} = kx = k \Delta x$

$$F_{sp} = T + F_g \sin \theta$$

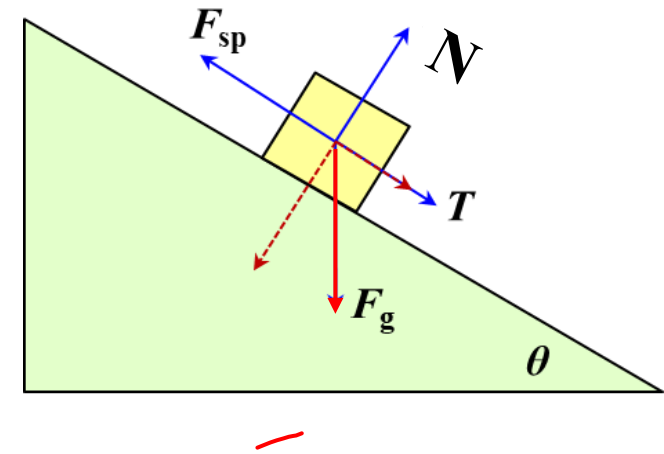
$$kx = T + mg \sin \theta$$

$x = \text{ ______ } m$

$$\Delta x = x - x_0$$

extended $\rightarrow x > x_0$

compressed $\rightarrow x < x_0$



$$\sum F_y = N + F_g \cos \theta = 0$$

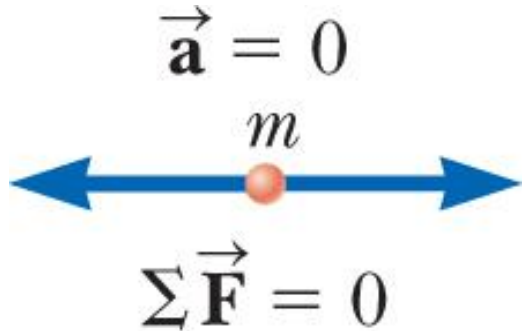
$$N = F_g \cos \theta = mg \cos \theta$$

$$N_{max} = ? \text{ \& } N_{min} = ??$$

Summary: Applying Newton's Laws

Equilibrium Problems

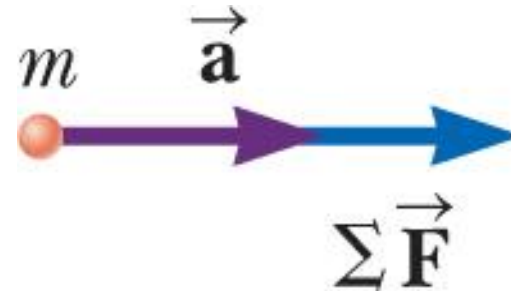
$$\sum \vec{\mathbf{F}} = 0$$



Find out force or mass.

Dynamics Problems

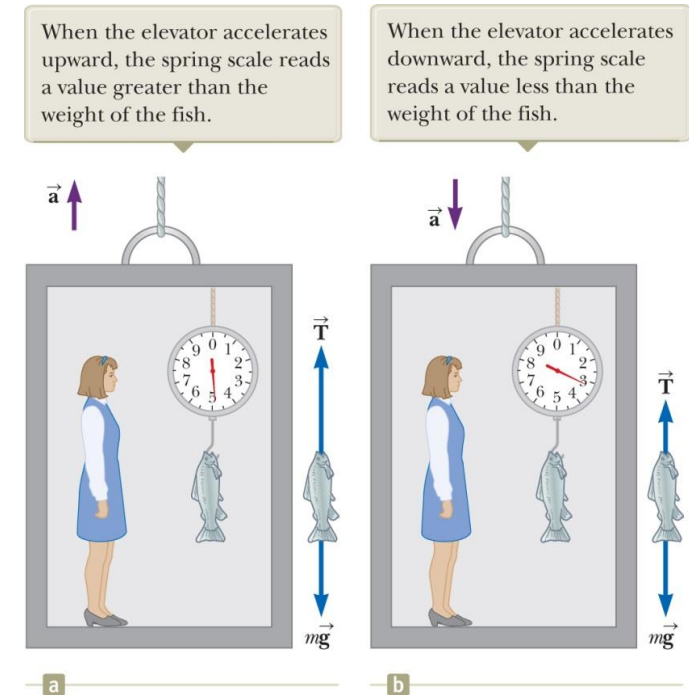
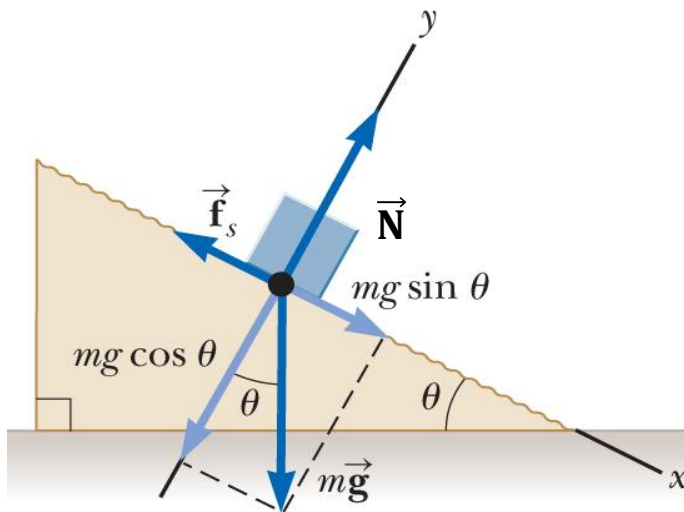
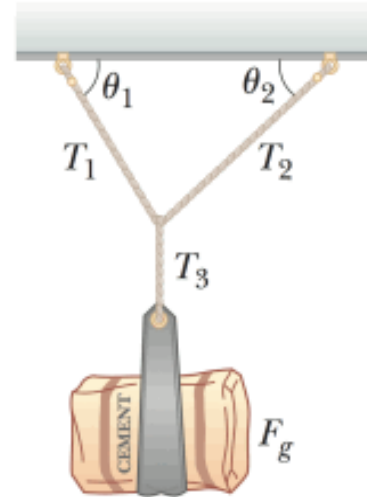
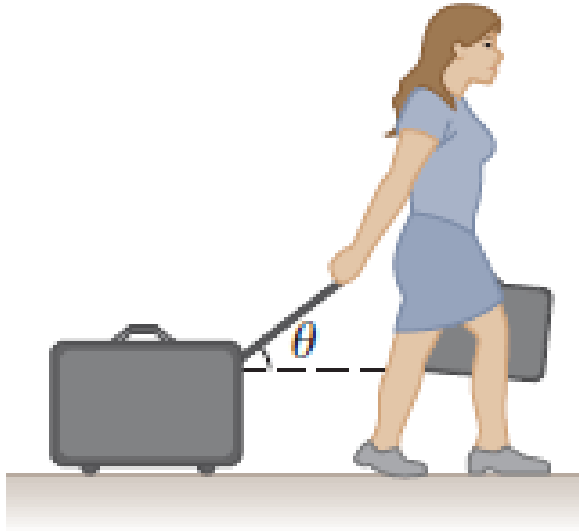
$$\sum \vec{\mathbf{F}} = m\vec{\mathbf{a}}$$



Use **kinematics** to find velocities and positions.

Summary: Applying Newton's Laws

■ Single Object Problems



$$\sum \vec{F} = m\vec{a}$$

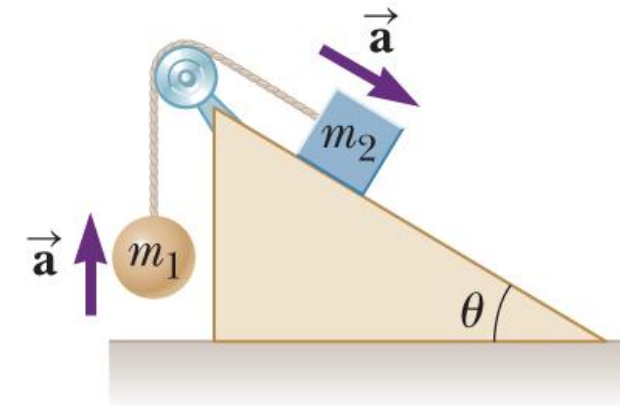
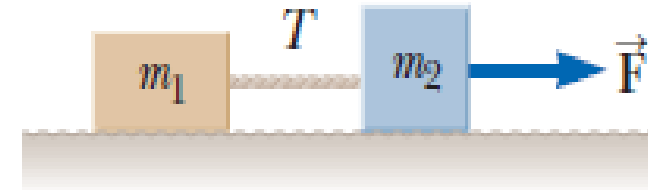
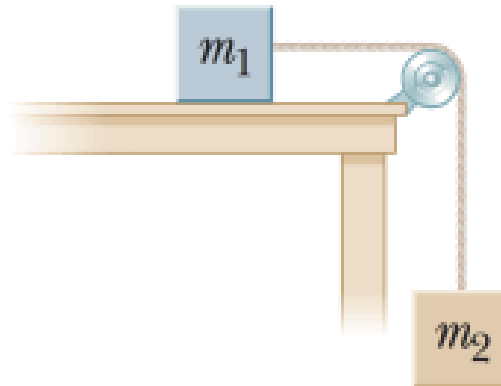
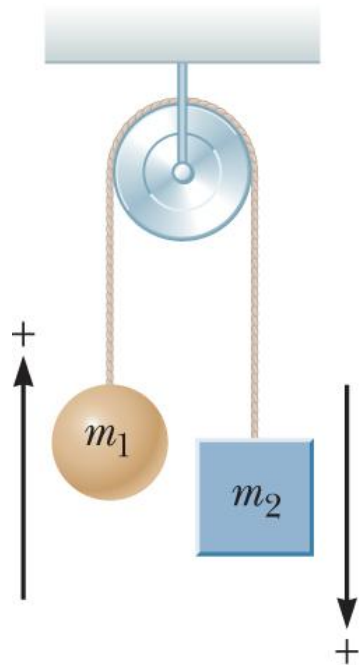
+

kinematics equations

Summary: Applying Newton's Laws

■ Two connected object Problems

- (1) Draw a free-body diagram for **each object**;
- (2) Tension force has the same magnitude throughout the rope;
- (3) Move together $\rightarrow$ the magnitude of acceleration is the same.



Summary: Applying Newton's Laws

■ Newton's 1st & 2nd Laws

- Draw and interpret a free-body diagram.
- Describe the consequences of no force or balanced forces on an object.
- Write **Newton's 2nd Law** for a physical scenario.
- Use **Newton's 2nd Law** to solve a problem
 - With **balanced** forces.
 - With **unbalanced** forces.
- Use Newton's Laws to predict the motion of an object.

■ More forces & Newton's 3rd Law

- Use **Newton's 3rd Law** to identify force pairs.
- Solve problems involving static friction, kinetic friction, gravity, tension, and normal force.